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Lab Sheet-09

Smart Digital Expense Tracker with Budget Prediction

BCA I Semester (Session 2026-2027)

Objective:

To develop a smart expense management system that helps users record daily expenses, categorize spending, monitor monthly budgets, and predict whether the monthly budget will be exceeded.

Requirements:

- C++ programming environment
- Text editor or IDE

Problem Statement:

Many students spend money daily but do not track their expenses. This project provides an easy way to manage expenses and alerts users before they exceed their monthly budget.

Modules

1. User Registration
2. Login System
3. Add Daily Expense
4. View Expense History
5. Category-wise Expense
6. Monthly Expense Report
7. Budget Setting
8. Budget Warning
9. Expense Prediction
10. Search Expense
11. Delete Expense
12. File Saving

Concepts Used

- Variables

- Loops
- Arrays
- Structures
- Functions
- File Handling
- String
- Searching
- Sorting

Future Enhancement

- AI-based spending prediction
- Mobile App
- QR Bill Scanner
- Cloud Backup

Learning Outcomes

Students will learn

- File Handling
- Real-world problem solving
- Menu-driven programming
- Data management
- Report generation

Instructions:

- Write comments to make your programs easy to understand and maintain.
- Use meaningful and descriptive variable names that clearly indicate their purpose.
- Follow proper code indentation and formatting for better readability.
- Test each program with different input values before submission.
- Ensure that your programs compile and execute without any errors.

Guidelines:

- Use clear, simple, and consistent C++ syntax throughout the programs.

- Declare and use appropriate variables and data types based on the problem requirements.
- Handle user input and output efficiently using cin and cout.
- Apply suitable programming constructs such as conditional statements, loops, functions, arrays, or classes wherever required.
- Keep the program logic simple, modular, and easy to understand.
- Display meaningful messages and outputs to improve user interaction.
- Follow standard C++ programming practices and coding conventions.

Submission:

- Submit the source code (.cpp) file for each program.
- Include a brief project/lab report containing:
  - Objective of the program
  - Algorithm or logic used
  - Source code
  - Sample input and output
  - Conclusion
- Ensure all programs are compiled, tested, and free from syntax and logical errors before submission.
- Submit all required files before the specified deadline.